

UMBC SUAS Safety Checklist 2025-2026

While no batteries are plugged in:

- ☐ FAA & AMA numbers are on plane
- ☐ Remote ID module is attached
- ☐ Pilot has AMA, TRUST, and FAA registration on h
- ☐ Propulsion & system battery is $\geq 90\%$ charge
- ☐ No obvious damage to fuselage, lift surfaces, or propellers
- ☐ All bolts + nuts are secure (PLEASE check)
 - ☐ Bolts attaching wing into fuselage
 - ☐ Bolts for landing gear to fuselage
 - ☐ Bolts for landing gear to landing gear wheels
 - ☐ Bolts for nose cone
 - ☐ Bolts for rear landing gear
 - ☐ Bolts for fuselage to vertical stabilizer 3-D printed connection
 - ☐ Bolts for vertical stabilizers
 - ☐ Bolts for horizontal stabilizers
- ☐ All servos are secure
 - ☐ Two @ ailerons
 - ☐ Two @ flaps
 - ☐ One @ elevator
 - ☐ One @ each rudder
 - ☐ One @ ground steering
- ☐ All avionics are secure
 - ☐ All servo wires are connected to the flight controller
 - ☐ Ailerons
 - ☐ Flaps
 - ☐ Elevator
 - ☐ Rudder
 - ☐ Ground Steering
 - ☐ All ESC wires are connected to the flight controller
 - ☐ GPS is connected to the FC
 - ☐ Lidar is connected to the FC
 - ☐ Airspeed Sensor is connected to FC
 - ☐ Raspberry Pi is connected to the FC
- ☐ Propellers are removed
- ☐ Propulsion & system battery is secure

- ☐ Shaking and moving the airplane crazy style does not result in any rattling
- ☐ Transmitter & GCS are active
- ☐ No low battery warnings on transmitter
- ☐ Primary GCS computer is above 50% battery
- ☐ Healthy connection between GCS and UAV
- ☐ Pitot tube is covered

Next, plug in system battery (NOT propulsion yet):

- ☐ Airplane held stationary for 30s after battery installation
- ☐ Test all control surfaces
 - ☐ Ailerons have symmetrical deflections @ minimum $\pm 15^\circ$
 - ☐ Elevator have symmetrical deflections @ minimum $\pm 20^\circ$
 - ☐ Both rudders have symmetrical deflections @ minimum $\pm 10^\circ$
 - ☐ Flaps have symmetrical deflections @ minimum $\pm 30^\circ$
 - ☐ Ground Steering has deflections @ maximum 10 deg
- ☐ Pilot can recall all switches correctly
 - ☐ Control surfaces (RETA)
 - ☐ MODE switches (Manual, FBWA/AUTOTUNE, Auto)
 - ☐ Flap switches
 - ☐ RTL switch
 - ☐ Camera switch
- ☐ Activating FBWA causes correct counter deflections when airplane is externally disturbed

With propellers off, connect the propulsion battery:

- ☐ Ensure motors spin in the correct direction.
 - ☐ Left motor CW
 - ☐ Right motor CCW
- ☐ Ensure ESCs are calibrated (is there dead space in thrust?)
- ☐ 0 to 100% throttle works
- ☐ Disarm cuts throttle immediately

Test failsafe in case of RC signal loss:

- ☐ Increase throttle to 50%
- ☐ Turn off the transmitter
- ☐ Automatic RTH/RTL After 15 Seconds of Comms Loss
- ☐ Automatic Flight Termination After 90 Seconds of Comms Loss

With propulsion disarmed, disconnect the propulsion batteries and attach the propellers:

- ☐ Propeller directions correct
- ☐ CG test passed
- ☐ Top fuselage lid is attached

With propulsion disarmed, connect the propulsion batteries:

- ☐ Aircraft directed in a safe direction
- ☐ Restraints situated behind the prop arc
- ☐ 0 to 100% throttle works while pointing the fuselage upwards. Ensure 1:1 thrust to weight ratio
- ☐ Disarm cuts throttle immediately
- ☐ Control surfaces move even with full throttle

Place airplane on runway and remove pitot tube cover:

- ☐ Taxi can be done straight
- ☐ Ground steering provides usable steering
- ☐ Airspeed is obtained
- ☐ No arming warnings are present

Set the airplane ready for flight. Ensure the pilot:

- ☐ Is awake and mobile
- ☐ Does not show symptoms of fatigue/dehydration
- ☐ Has nothing obstructing controls
- ☐ Enough finger sensitivity
- ☐ Is not shivering
- ☐ Is goated (leave unchecked lol)

Before and during flight, remind the pilot of the following:

- ☐ "Stay well over the height of trees except when coming for a landing"
- ☐ "Fly over 200ft when flying in autonomous modes"
- ☐ "Wear sunglasses if sunny"
- ☐ "Do not go too far out without warnings"

Remind others of the following:

- ☐ Pilot is the furthest forward
- ☐ GCS operator and primary camera-person are next to pilot
- ☐ Everyone (except GCS, primary camera-person, and pilot) are behind fences
- ☐ GCS view is being recorded
- ☐ Keep track of flight time

Date:

Session # and Flight #:

Safety Pilot:

GCS Operator: